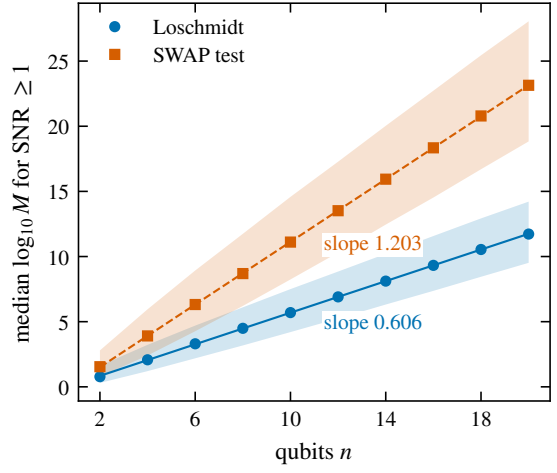
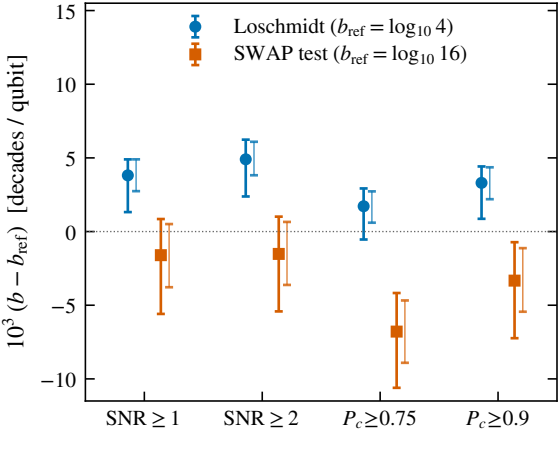
(a)

$$F(\theta) = \prod_j \cos^2(\theta_j/2), \ \text{target } |0^n\rangle; \quad F_\pm = F(\theta \pm \tfrac{\pi}{2} e_k), \ g = \partial_k(1-F) = (F_- - F_+)/2$$

Loschmidt readout
each shot $\sim$ Bernoulli($F_\pm$)
$\hat{g}_{\mathrm{LE}} = (K_- - K_+)/(2M)$
per-shot variance of $\hat{F}$: $F(1-F)$

SWAP test readout
each shot $\sim$ Bernoulli($(1+F_\pm)/2$)
$\hat{g}_{\mathrm{SW}} = (K_- - K_+)/M$
per-shot variance of $\hat{F}$: $1-F^2$

(b)



(c)